

William Liu

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Motivated high school senior with strong problem-solving ability, leadership experience, and a passion for STEM. Enjoys solving challenging problems, building creative projects, exploring advanced maths and sciences, and promoting STEM within the community. Aspiring to become a scientist or engineer to help create a better shared future.

Education

Ronald Reagan High School - 12th grade, expected graduation May 2027

- **Class Rank:** 2/850 (Unweighted GPA 99.2/100, Weighted GPA 108.2/100)
- **Test Scores:** SAT: 1600 | AIME: 6 | AMC 12: 108
- **Relevant Coursework:** AP Calculus BC, AP Physics C: Mechanics, AP Chemistry, AP Statistics

Experience

Welch Summer Scholar, Henkelman Lab, UT Austin

June 2026 - Present

- Engineered a Python-based materials screening pipeline using machine learning interatomic potentials (MLIPs) to accelerate discovery of electrode materials
- Screened 150,000+ candidate materials using MLIPs, and used results to train a generative AI model to propose novel electrode materials for lithium extraction
- Preparing a publication documenting novel materials discovery methodology and results

Independent Developer

August 2023 - Present

- Created many projects to explore the capabilities of programming and mathematics
- Built a Fourier Transform simulator, a bell schedule utility, and a Valentine's Day applet, among other projects
- Designed and deployed a customized personal website with Svelte to showcase portfolio of projects and technical skills

Reagan Mu Alpha Theta

VP of Academics (April 2026 - Present), Inducted Member (October 2023 - Present)

- Led academic programming for 100+ members including math competitions, test prep sessions, and math-related events
- Founded a math team and designed curriculum to prepare students for competition mathematics, bridging gap between standard curriculum and AIME-level content

Reagan Orchestra

Violinist (August 2023 - Present), Librarian (May 2025 - May 2026)

- Auditioned into varsity Chamber Orchestra and All-Region Orchestra multiple times
- Performed in many public events, including school musicals, seasonal concerts, and community outreach performances
- Managed copying, inventorying, organizing, and distributing music as a former Librarian

Skills

- **Programming Languages:** Python, Java, JavaScript, HTML/CSS, C++
- **Programming Frameworks:** Pandas, NumPy, PyTorch, Svelte, matplotlib, TailwindCSS, SQLite
- **Other Interests:** Playing violin, running, swimming, reading, talking about public transit, watching NFL games

Awards

- American Invitational Mathematics Examination (AIME) Qualifier (2024-2026): Only school qualifier, first freshman AIME qualifier in school history
- American Mathematics Competitions (AMC) 1st Place, Reagan High School (2023-2025)
- Stanford Math Tournament 5th place Team Round with the San Antonio Math Circle (2026)
- USA Mathematical Talent Search (USAMTS) Honorable Mention (2026)
- USA Computing Olympiad (USACO) Silver Division
- Student Volunteer Service Award - Gold Level (2025, 2026)
- Texas State Solo & Ensemble Competition qualifier (2024-2026)